



An hybrid machine learning and improved social spider optimization based clustering and routing protocol for wireless sensor network

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Abstract

Wireless Sensor Networks (WSNs) monitor and gather environmental data by interconnecting numerous sensor nodes spread across space via wireless communication. These nodes operate on battery power, which depletes over time, thereby limiting the network's operational lifespan. This energy constraint significantly impacts the overall longevity of the network. The primary focus of the work is to reduce energy consumption and increase the network's lifespan. To this end, WSNs currently make extensive use of routing and clustering algorithms. This work selects an optimal Cluster Head (CH) from a set of nodes using a combination of hybrid Ant Colony Optimization (ACO) and the Improved Social Spider Cluster Optimization Algorithm (ISSOA). The selection process takes into account a number of variables, such as the nodes' residual energy, their degree and centrality, their proximity to nearby nodes, and their distance from the Base Station (BS). Furthermore, we use an optics-inspired optimization (OIO) algorithm to determine the path between the chosen CH and the BS. In order to ensure effective data transmission throughout the network, this algorithm optimizes the path based on variables including distance, node degrees, and the residual energy of nodes along the route. The simulation results show that the proposed ACO-ISSOA method significantly improves several Quality of Service (QoS) parameters. The proposed method works better than other algorithms like Low Energy Adaptive Central Hierarchical Clustering (LEACH-C), Multiple-Weight LEACH (MW-LEACH), Hybrid Genetic Algorithm with Particle Swarm Optimization (GA-PSO), and ACO-based Hierarchical Clustering (ACOHIC). The ACO-ISSOA protocol improves network lifetime (5700 rounds), throughput (99%), PDR (99.5%), energy consumption (56 mJ), and execution time (45 s) for CH selection. When compared to other algorithms, the hybrid ACO-ISSOA algorithm outperforms them.

Keywords Wireless sensor network · Clustering · Cluster head selection · Routing · Ant colony optimization, improved social spider optimization · Optics inspired optimization

1 Introduction

The advancement of integrated circuit design and wireless communication technologies has enabled the creation of WSNs, with sensor nodes serving as their fundamental core components [1]. Several industries, including smart homes, traffic management, healthcare, and disaster monitoring, have recently adopted WSNs. Numerous small, low-power sensor nodes, scattered throughout an environment, form a WSN and perform a variety of applications (hence the name "heterogeneous") or the same task as a group (i.e., homogeneous). WSNs typically have severe limitations on their network life, node processing capabilities, memory, data transfer restrictions, scalability, and failure tolerance [2–4]. Every node in a WSN is able to sense the environment and transfer data to a base station for further analysis.

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Each node consists of one or more sensor units, a transceiver unit, a computing unit, and a limited power supply unit [5]. Because it is challenging to recharge the sensor network's power sources, often found in hostile environments, a power routing protocol is essential to prolonging its life. Routing is the most common approach to solving WSN issues [6]. Consequently, we group the sensor nodes to enable efficient data collection. We should select a CH based on a few pre-determined standards that enhance network efficiency [7].

Artificial intelligence, data security, smart city infrastructure, and natural environment monitoring are just a few of the numerous applications for WSNs. These networks consist of wireless sensors that collect data and send it to a base station. The BS analyses the collected data to make accurate decisions based on the current circumstances. However, WSNs face significant challenges due to energy constraints, as sensor nodes typically use micro-battery power sources. In harsh or hazardous environments, recharging or replacing these batteries can be challenging. Furthermore, certain sensor nodes function autonomously for long periods of time, resulting in significant energy use and swift depletion, commonly referred to as the "energy hole" problem. This problem can interrupt signal links, hinder efficient data transmission to the BS and complicate WSN networking. Consequently, there is a pressing need to explore and devise strategies to enhance energy efficiency in WSNs to ensure stable network operations.

Two main factors significantly impact the energy efficiency of WSNs in the present moment: (1) the energy used by sensor nodes to collect data in their environment, and (2) the energy required during the transfer of data between nodes and a BS. In WSN, using hierarchical routing protocols can effectively resolve these issues. Nevertheless, these protocols possess certain limitations that necessitate careful consideration. Specifically, WSNs recognize the clustering problem as NP-hard, making it exceedingly challenging to find the optimal set CHs. In addition, CHs face a higher level of data aggregation from their child nodes, leading to an increased workload and faster depletion of energy. This ultimately reduces their operational lifespan. Also, trying to make CHs use the same amount of energy may cause an uneven distribution, with CHs closer to the BS using up energy faster and being more likely to experience the "hot zone" effect. In addition, hierarchical routing protocols can unintentionally create lengthy forwarding paths, which can decrease the longevity of the network. Therefore, it is crucial to develop a hierarchical routing protocol that effectively chooses CHs and optimises the transmission paths for forwarding data. The goal of this approach is to reduce the amount of network communication required, achieve equilibrium, and improve the network's overall lifespan [8].

Recently, there has been a growing interest in using Swarm Intelligence (SI) algorithms for clustering techniques

in hierarchical routing protocols. These algorithms provide more efficient methods for selecting CHs and determining near-optimal routes within reasonable timeframes. The use of SI algorithms improves CH selection and path determination by overcoming the limitations of individual SI algorithms. This hybrid methodology enhances the network perception capabilities of nodes, achieves a more optimal equilibrium between exploration and exploitation, and yields a more logical and efficient selection of CH and path. SI-based designs derive inspiration from the collective behaviours observed in creatures such as grey wolves, fish, and bats. These behaviours are mathematically simulated to enhance network efficiency. Various SI techniques, such as Grey Wolf Optimization (GWO), Butterfly Optimization Algorithm (BOA), Particle Swarm Optimization (PSO), Marine Predator Algorithm (MPA), Artificial Bee Colony (ABC) and Cuckoo Search Algorithm (CSA), were used to optimize CH selection and routing paths in WSNs. However, relying solely on a single SI algorithm can pose challenges such as local optima and limited global exploration, which may not effectively lead to identifying optimal CHs [9, 10]. Building upon earlier SI algorithms, this study centres on a novel two-level hybrid intelligent optimization algorithm for SI. In the initial stage, the algorithm determines the optimal CH using a hybrid of ACO and ISSCOA algorithms. Subsequently, in the second stage, the OIO algorithm is employed for routing selection, effectively resolving hotspot problems and enhancing the longevity of network services.

In this work, the random placement of nodes in WSNs often leads to energy hotspots and inefficient data forwarding paths. Additionally, using a single swarm intelligence algorithm often results in local optima, causing higher energy consumption and a shorter network lifespan. To address these challenges, this research proposes a hierarchical cluster-based routing protocol using a two-tier hybrid SI method. The suggested method combines a hybrid ACO method with an improved clustering method that uses ISSOA to get around the problems that come with cluster-based inference and metaheuristics, which are often slow. Additionally, this work proposes an improved routing protocol based on OIO for WSNs. ACO algorithms mimic ants' decentralized behaviour, making them ideal for WSNs in which sensor nodes operate independently and must make autonomous decisions without centralized oversight. ACO algorithms help to extend network longevity by clustering nodes and implementing energy-efficient routing techniques that distribute energy consumption evenly among nodes.

By leveraging social spider-like behaviors, ISSOA improves CH selection, making it easier to identify nodes with significant centrality and connectivity within the network. ISSOA is distributed, allowing sensor nodes to select CHs autonomously based on local information and interactions with neighbouring nodes. This decentralized approach

minimizes the need for centralized control and communication overhead, making it appropriate for large-scale WSN deployments. OIO mimics the behavior of light rays, which naturally seek the most efficient paths through a medium. Using this behaviour, these algorithms can efficiently select routing paths in WSNs, reducing latency and packet loss. The major contributions of the proposed work are summarized as follows:

- Two distinct hybrid SI algorithms address clustering and routing problems, effectively tackling challenges related to optimal search.
- The proposed clustering algorithm, based on a hybrid of ACO and ISSOA, achieves a balanced exploration–exploitation trade-off. This approach optimally selects CHs within the network, aiming to extend both network lifetime and energy stability.
- The approach employs a robust fitness function that assesses multiple objectives, including network coverage, residual energy, communication cost, proximity to the selection of CHs, and network clustering.
- We also use the OIO algorithm to determine the best path for data transmission between the chosen CH and the sink node. We employ a route alternation strategy to explore viable data transmission routes, thereby mitigating the risk of route failure.
- We conducted simulation experiments to assess its efficiency against existing algorithms, using evaluation metrics such as throughput, energy consumption, PDR, NLT, jitter, buffer occupancy, and PLR across various sensor nodes and rounds.

The remainder of the article is divided into the following sections: The following section provides an overview of the current CH selection and routing methods. Section 3 presents the suggested cluster head selection method, routing algorithm, and potential function. Section 4 analyses the proposed work using several current conventional and population-based algorithms. Section 5 concludes the work.

2 Literature survey

Juana Matos et al. (2022) suggested a metadata review of the current situation for enhancing wireless sensor networks. The study displays and discusses the most prevalent WSN issues. This study employs the multi-objective (MOO) optimization method as the most suitable technique to demonstrate complex multi-objective functions that comprise numerous WSN problems. Power, coverage, and aggregation are the subjects that appear most frequently in WSN articles, followed by the subjects listed above. Multi-objective optimization (MOO) is the best technique because

many variables might impact a WSN's quality of service. We provide examples of applications and assess each method's applicability to WSN issues [11].

Preeth et al. (2018) developed the Fuzzy-Based Energy-Efficiency Aggregation and Immune-Inspired Routing (FEEC-IIR) protocol. The protocol applies an immunity-inspired optimization technique to enhance the reliability of data delivery for routing. We gathered data from the real-time data streaming scenario and entered it into MATLAB. In order to compare it to other routing protocols [12], we plot performance results against QoS metrics such as packet delivery rate, packet drop, bandwidth, network lifetime, end-to-end delay, channel load, jitter, rate of bit errors, buffer occupancy, and power usage.

Celestine Ewende et al. (2020) developed a method to increase the network's lifespan by lowering the energy consumption of sensors in the Internet of Things. This study identifies the IoT network's most suitable group header (CH) to minimize power requirements. The suggested study employs the Whale Optimization Method (WOA), a hybrid metaheuristic algorithm, and Simulated Annealing (SA). In the future, many additional performance criteria, such as link lifetime, delay, etc., can be considered to determine the optimum CH to increase power utilization further [13]. The Neuro-fuzzy Emperor Penguin Optimization (NF-EPO) method was created by Preeth et al. (2020) as a smart way to design paths that use less energy for sink-based mobile WSNs. The Effective Orientation Algorithm for Emperor Penguin Optimization (EPO) ultimately determines the meeting spots and transit path for the portable dock. Comparing the suggested method to other current routing systems, the simulation results demonstrate that it offers excellent performance [14].

Arjun K. P. et al. (2021) developed a neural assembly-based energy adjustment using the IoT EAANFC-MR algorithm assisted by WSN. Experts suggest that EAANFC-MR consists of two main phases: the addition and implementation of several EAANFC-based stalls. A clustering technique based on EAANFC selects CHs with residual energy (RE), distance, and nodal degrees. We employ the QOBFO algorithm as a multiple-trajectory method to determine the optimal routes to the destination. The comparison showed that the EAANFC-MR model had higher power loss, network lifespan, PDR, package loss, transmission efficiency, and delay [15].

SR Deepa et al. (2020) created an algorithm for Bacterial Foraging Search Optimization (BFO). They contrast the performance of BFO with that of LEACH and PSO. Enhancing WSN cluster formation and reducing the number of surviving nodes should extend the network's lifespan. Abandoned nodes, which use more energy to relay data to the base station before eventually dying, shorten the network's lifespan. BFO prevents the creation of abandoned nodes [16].

MoFIS-BFO was created by Elaheh Moharamkhani and his colleagues in 2021. It is a flexible fuzzy inference system (MoFIS) that includes a bacterial foraging optimization-based fuzzy knowledge metaheuristic model (BFO). The MoFIS-BFO model uses moFIS to estimate the likelihood of each node becoming a Cluster Head (CH) based on a variety of variables, including temperature, residual energy, distance to neighbors, and distance to the base station. The simulations demonstrate that the proposed MoFIS-BFO protocol outperforms current methods in terms of improving data packet delivery rates, making the network last longer, and sending more packets to the base station. Finally, it successfully transmits information to the base station.

Abhilash Singh et al. (2021) presented nature-inspired optimization strategies for the best WSN coverage. The article briefly covered the classification of optimization methods and the issues with WSNs. In contrast to other methods, the LO condition requires fewer trials to attain optimal coverage. This is because there are many local maximums with an excellent fitness value, making it practically impossible to miss the local maximums. There is still much room for exploration and use of LO in multi-objective situations, even though it performs better than other optimization techniques [18].

Kalavajunta Aravind et al. (2022) used the best CH for the Internet of Things Dingo's self-adaptive brownian motion optimizer (SDO-BM) to think about many things, like power, distance, delay, global traffic, quality of service (QoS), and security. Group-based routing has some advantages over wired systems in the Internet of Things, including simplicity of usage, reduced delivery costs, decreased failure risk, and enhanced mobility [19].

Asha Jerlin Manuel et al. (2020) detail these older models' research gaps. The group's primary focus is managing power factors in WSN. This study covers current procedures and algorithms. We divide the parameters, optimization, and technique of these protocols into different categories. In terms of grouping features, both primary and secondary criteria are considered. We have now assessed and examined each class of procedures in relation to the given parameters. This review offers another viewpoint and a starting point for investigation approaches by looking into classified methods to clearly explain the flaws of the protocols, with the ultimate goal of providing valuable data and inspiring investigators [20].

Aditya Tandon et al. (2021) proposed an IoT routing and energy conservation strategy. A hybrid algorithm inspired by neighbor interactions combines salp swarm optimization and moth search to determine the routing path. Performance analysis showed that the proposed method used 30 mJ of energy, transmitted 0.99 Mbps, had a 2% BER, had less than 2% packet drop, and maintained 2% buffer occupancy. The network could handle 5600 transmissions, with an average

transmission delay of 3.2 s. BiHCLR proved superior to existing technologies [21].

Al-Hajjar et al. (2023) reviewed modern IoT-integrated ML epilepsy detection applications. The study conducted a systematic analysis of selected literature on epileptic seizure detection techniques, including data sources, ML classifiers, and feature extraction techniques. The research looked at a number of different feature extraction methods and found that SVM, RF, KNN, LR, DT, NB, and ANN were the best at classifying data [22]. Mohammed et al. (2021) developed a sound sensitivity diagnostic framework using IoT and cloud-based facial expression recognition systems. They developed a cognitive IoT hardware setup and software to monitor sound-sensitive people. To monitor this, they used cameras and image processing methods, including CNN software for real-time facial emotion tracking in videos. CNN outperformed fuzzy algorithms by 80%, despite longer training times [23]. Al-Hajjar et al. (2023) developed a feature importance method using the Extra Tree Classifier (ETC) and Extreme Gradient Boosting Classifier. To evaluate their efficacy, they compared the two methods using Bonn University experimental data. The ETC approach is less accurate than XGBoost [24].

Raheem et al. (2022) used supervised ML algorithms to create an accurate prediction model for early-stage diabetes in humans to prevent severe morbidity and mortality. Their model extracts hidden insights from Kaggle ML diabetes data. This study develops a web service that allows users to enter data and get diabetes diagnosis predictions, which benefits the healthcare industry. Decision Forest outperformed other ML strategies in classification [25].

Abdulzahra et al. (2022) used fuzzy C-means to cluster WSN-based IoT systems. First, FCM clusters and reduces energy consumption within each cluster before choosing the best CH. We propose using an energy threshold to predict CH dynamicity based on current energy levels, unlike previous studies that suggested periodically replacing CHs for dynamic clustering. This method prolongs sensor network life [26]. Abdulzahra et al. (2023) proposed the Energy-Efficient Fuzzy-Based Unequal Clustering with Sleep Scheduling (EFUCSS) protocol for WSN-based IoT applications. This protocol optimizes clustering, scheduling, and data transmission to extend network life and reduce energy use. Forming unequal clusters with FCM reduces data transmission distances, saving energy. A fuzzy logic system selects CHs. Compared to existing methods, the proposed method increases remaining energy (26.72–23.4%) and network lifespan (39.58%–408.13%) [27].

Al-Qurabat et al. (2021) created a lossless, lightweight compression algorithm using differential encoding and Huffman techniques for Internet of Things (IoT) nodes that monitor environmental features but don't have a lot of memory or computing power. Temperature measurements demonstrate

superior performance compared to conventional WSN methods, resulting in significant reductions in data transmission volume and energy savings. However, the proposed system does not achieve its theoretical maximum efficiency [28].

Al-Qurabat et al. (2022) suggested a lossless, lightweight compression algorithm based on DE and Huffman techniques for IoT sensor nodes that monitor environmental features but don't have a lot of memory or computing power. Given a general understanding of monitored features, classical Huffman coding can describe them across periods and locations without creating novel ad hoc algorithms. Although the proposed system does not reach theoretical efficiency, temperature measurements outperform conventional WSN methods [29]. Al-Qurabat et al. (2022) introduced a PIP-DA approach for WSNs to reduce redundant data before sink transmission. Intel Berkeley Research Lab's (IBRL) data assessed the method's efficiency. Compared to PFF and ATP [30], PIP-DA reduces sensor node data overhead by 1.25% and energy consumption by 93% [30].

Al-Qurabat et al. (2020) provide a thorough overview of periodic WSNs (PWSNs), including an examination of their applications and challenges. Nodes in PWSNs collaborate to monitor their environment and facilitate human-network interaction. The proposed work succinctly defines PWSNs and describes their characteristics in great detail. Their widespread use in various applications, particularly in IoT, is becoming more widely recognized due to their importance as data sources [31].

Al-Qurabat et al. (2022) suggested a data collection strategy that focuses on extracting critical extrema points to extend the lifespan of WSN. Instead of transmitting all collected measurements at the end of each period, their approach involves transmitting only the most important extrema measurements from the sensor nodes. They tested their method on various real-world datasets and compared it to the existing protocols. When compared to existing protocols, the results showed significant reductions in remaining data (up to 95%), data sent (up to 80%), and energy consumption (up to 77%). [32].

Al-Qurabat et al. (2022) introduced the Extrema Points Extraction (DAEP) method for WSNs, designed to significantly reduce energy consumption per node and extend the overall lifespan of WSNs. They conducted multiple simulation experiments using real sensor data from Intel Berkeley laboratories to assess the effectiveness of their approach and compare it with existing literature. The results show that the proposed method can reduce the load on sensor nodes by 69–80% and energy consumption by 73–77% while maintaining an acceptable level of accuracy when compared to previous works [33].

Al-Qurabat et al. (2022) created an energy-saving clustering algorithm (ESCA) to save energy and preserve network life. The algorithm relies on distributed CHs and centralized

cluster construction. Throughout the operation, a centralized K-means algorithm determines static clusters. Using a custom Python simulator, we compared the proposed method to MOFCA and IGHND clustering. The ESCA solves energy utilization issues and extends network lifespans [34]. Al-Qurabat et al. (2023) proposed a WSN-based FCM-based IoT clustering method. To choose the best CH, FCM forms clusters and reduces energy consumption within each cluster. This study proposes using an energy threshold to predict CH evolution based on current energy levels, which will extend the sensor network's lifespan [35]. Traditional dynamic clustering methods update CHs on a regular basis.

Al-Qurabat et al. (2023) conducted a thorough analysis of various clustering techniques, including their objectives, characteristics, and other relevant factors. Clustering techniques are classified and evaluated using cluster features, CH characteristics, and clustering procedures, according to the study. [36]. Al-Qurabat et al. (2023) categorize and evaluate existing prediction-based data reduction strategies designed for WSN. Their main contribution is a methodical way to choose prediction models in WSNs that takes into account the limitations of the network, the features of the prediction technique, and the features of the data that has been collected [37].

Fahimeh Aghaei et al. (2019) introduced the Multi-rate Multicast Scheduling (MMS) algorithm for single-cell multi-channel cognitive radio wireless mesh networks. This algorithm leverages multi-radio and multi-rate technologies. Research indicates that multi-radio nodes reduce multicast periods and enhance network throughput. Their approach minimizes transmissions and takes advantage of high transmission rates [38]. Fahimeh Aghaei et al. (2020) proposed the cross-layer algorithm "Multicast Routing, Channel Selection, Scheduling, and Call Admission Control (MRCSC)." This algorithm uses wireless broadcast, channel/radio diversity, and spatial reuse to construct multicast trees with specific bandwidth requirements via intra- and inter-cell relaying. Extensive simulations showed that MRCSC improves transmissions, network stability, and multicast duration [39].

Pratik Thantharate et al. (2024) proposed an innovative joint optimization model for connecting multiple aerial BS UAVs for charging. The model optimizes the placement of charging stations and UAV trajectories for improved flight time and energy efficiency. Effective charging strategies make use of UAV mobility patterns, fully utilize infrastructure with balanced distribution, and strategically leverage existing BSs before deploying dedicated charging assets. Global optimization enhances UAV battery life and efficiency [40]. To balance sensor node energy consumption, S. Reddy et al. (2023) introduced spider and fly optimization. Reusing information from source nodes to the sink reduces the number of routing messages and energy waste

in the protocol. This routing protocol selects the best route. To determine the optimal CHs in each round, the proposed algorithm uses a variety of criteria, including node residual energy, inter-cluster distances to the sink, and cluster overlaps [41].

Supreet Singh et al. (2024) developed an attraction and repulsion strategy-based NMRA (ARNMRA) with self-adaptive properties to prevent solutions from becoming trapped in local optima. Factors such as node mobility, residual energy, and connection duration determine the CH selection in this protocol for mobile WSNs (MWSNs). The protocol's primary goal is to maximize system longevity by efficiently gathering and transmitting data from all sensors to the BS. Furthermore, this study broadens the scope of ARNMRA applications to include data collection aspects to increase network longevity and energy efficiency [42]. Table 1 evaluates WSN energy-effective clustering and routing algorithms.

2.1 Problem statement

In previous architectures, distributed clusters controlled all sensor nodes' operations and transmissions. Some networks, such as high- and low-node clusters, exhibit too much variation in their BS and CHs to achieve optimal energy performance for the model's changeable cluster size. However, the model's lengthy computations lead to delays and increased load. The earlier algorithms failed because they used too much energy. LEACH-C presented a cluster construction utilizing the simulation process. LEACH-C provides a pre-determined number of clusters, uniformly spaced, despite the overhead of cluster development in each round. Based on energy usage and data packet delivery, LEACH-C outperforms LEACH [52]. However, sensors only join a CH when they are close enough to the load-unbalancing distance. Researchers have proposed a PSO-based CH selection method [53]. A new fitness function was based on energy, intra-cluster distances, and node degree. This has greatly improved existing methods. Sensors only connect CHs based on distance, resulting in imbalanced clusters. GA-based clustering used a unique fitness value that included the distance and morbidity of the first and last nodes. Fitness function derivation does not consider energy [54]. The ACO algorithm determines the optimal route from each CH to BS. This method focuses on network longevity. More improvement is possible [55]. We selected the initial group of CH candidates from nodes with residual energy close to the cluster center's density. However, the model needed to generate an initial set of CH alternatives and use tested sensors for several application domains. We have implemented AHHO, SFO, Cross-Layer Protocol, and Levy Bat algorithms for WSN clustering and routing.

2.2 Major contribution

- The WSN creates cluster heads using ACO due to its stability and low computational complexity. The ISSOA chooses the CH based on residual energy, distance to neighbours, base station (BS), node density, and centrality. We adjust the node degree, distance, and residual energy parameters of the ACO to resolve ambiguous convergence times.
- Optics-inspired optimization (OIO) finds the shortest path between the CH and the BS. OIO-based networking uses a novel fitness function based on remaining energy, distance, and node degree. We also include an innovative mapping method and a robust encoding system.
- Energy-efficient CH selection and optimal data transfer paths extend the network's lifespan. By reducing node energy consumption during data packet transmission, base station message delivery increases.

3 Proposed methodology

The WSN sensor network rapidly increases sensing data. Transferring WSN sensor data directly to the gateway wastes energy and increases transmission errors. We must create a redundancy- and energy-efficient compression algorithm for WSNs. This study employs a hybrid approach combining Ant Colony Optimization (ACO) and the Improved Social Spider Optimization Algorithm (ISSOA) to select an optimal Cluster Head (CH) from a set of nodes. The selection process considers several factors, including the nodes' residual energy, degree and centrality, proximity to neighboring nodes, and distance from the Base Station (BS). Additionally, an optics-inspired optimization (OIO) algorithm is used to determine the optimal path between the selected CH and the BS. This algorithm ensures efficient data transmission across the network by optimizing the path based on variables such as distance, node degrees, and the residual energy of nodes along the route. To expedite computation, the ideal traversal path can be discovered using an optimization routing technique influenced by optics. Not only does the OIO technique prevent ineffective routing path communication, but it also ensures reliable data transmission. Figure 1 depicts the single-hop clustered WSN topology.

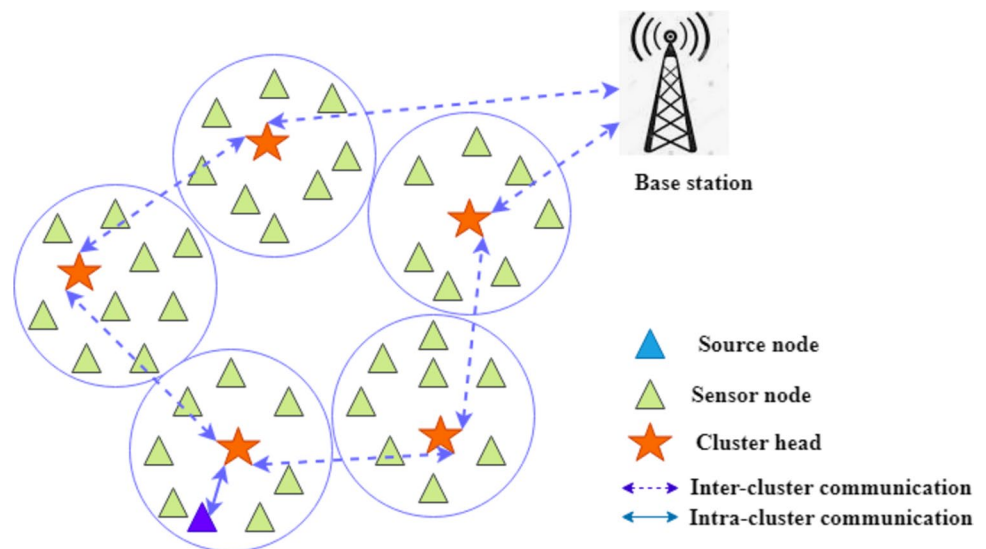
3.1 Network model

The network model specifies node placement and positioning to enhance data communication. The clustering process in WSNs divides individual nodes into groups known as clusters, overseen by a separate CH node, to ensure effective management. The following is a description of the energy model's various operational aspects:

Table 1 Performance assessment of several popular energy-efficient frameworks for WSNs

Reference No	Author	Algorithm	Advantages	Shortcomings
[14]	Sathya Lakshmi Preeth et al. (2020)	NF-EPO Algorithm	The experimental outcomes demonstrate that the suggested approach yields superior outcomes compared to conventional methods	The scalability of NF-EPO clustering in large-scale WSNs could be constrained by its computational complexity and convergence properties
[43]	Tabibi et al. (2018)	PSO algorithm	The recreation shows that the PSOBPS produced the best results	However, it increases the PLR during correlation by WRPBS
[44]	A. F. Raslan et al. (2021)	Improved Sunflower optimization algorithm	Higher number of active nodes. Enhanced network lifetime performance	Inadequate validation for real-time data stream management
[45]	M. Rathee et al. (2021)	ACO based routing Algorithm	Improved network lifespan with decreased latency	Future research is required for situations where determining weights is challenging or infeasible
[46]	S. Jagadeesh et al. (2022)	Oppositional AFS with Improved MFO	Maximize the Qos Metrics	Significant increase in the overhead of data transmission
[47]	A. Ojha et al. (2022)	Multi-objective GWO algorithm	Substantial enhancement in network lifespan	Improvements are needed for the fault detection method
[48]	S. S. Kalburgi et al. (2022)	Taylor-Spotted Hyena Optimization	Enhanced energy consumption	The difficulty in dealing with multi-objective optimization criteria
[49]	V. Cherappa et al. (2023)	Adaptive SFO-ECERP algorithm	Outstanding performance in QoS metrics such as PDR, network longevity and energy usage	High energy usage and latency issues
[50]	S. Chaurasia et al. (2023)	Meta Heuristic Cluster-Based Routing Protocol	Superior results in delay, average residual energy, throughput and PDR	Restrictions on investigating the best CH and route selection techniques
[51]	T. Luo et al. (2024)	Levy Chaotic PSO routing algorithm	Enhanced energy balance, evolutionary possibility, and routing efficiency	Difficulties in guaranteeing stability and dependable service quality for a range of applications

Fig. 1 Single-hop clustered WSN topology



Section 1: Node deployment strategy

Consider a network with 'N' sensor nodes evenly distributed across 'A'. For the specified area, the node density 'ρ' is defined as:

$$\rho = \frac{N}{A} \quad (1)$$

In the network model, a key parameter is node density (ρ), which represents the number of sensor nodes in the network area. This parameter is constant during optimization.

Section 2: Measure node distance

Node density and area characteristics can be used to calculate the average distance 'd' between nodes in a cluster:

$$d = \sqrt{\frac{A}{\pi \times N}} \quad (2)$$

Section 3: Cluster organization

The network's cluster count is:

$$C = \sqrt{\frac{N}{5 \times \pi \times R_N}} \quad (3)$$

The models above improve energy efficiency and cluster-based routing in WSNs in the ACO-ISSOA hybrid framework.

3.2 Energy model

The energy model is essential for understanding and optimizing network node energy consumption. It is widely used to calculate transmission and reception energy consumption. The energy model's components are listed below:

Section 1: Data transmission energy

$$E_{TX}(l, d) = (E_{elec} \times l) + (\epsilon_{amp} \times l \times d^2) \quad (4)$$

where: $E_{TX}(l, d)$ represents the energy consumed to transmit an 'l'-bit message over a distance 'd', set at 45 nJ/bit. E_{elec} denotes the energy dissipation per bit for transmitter/receiver simulation, fixed at 50 nJ/bit. ϵ_{amp} signifies the energy dissipation per bit per square meter for the amplifier, set at 0.0015 pJ/bit.

Section 2: Data reception energy

$$E_{RX}(l) = E_{elec} \times l \quad (5)$$

where $E_{RX}(l)$ denotes the energy consumed to receive an 'l'-bit message, set at 45 nJ/bit.

The proposed system functions under the following assumptions:

- The BS remains stationary in a predefined network location.
- The BS has an unlimited supply of energy and significant processing power.
- All nodes in the network are aware of the BS's location.
- Nodes are stationary and distributed at random throughout the network.
- Each node has a positioning system that allows it to determine its own location.

3.3 Cluster formation using ACO

The ACO algorithm is inspired by the foraging behavior of ants. Ant colonies communicate indirectly by leaving pheromone trails while searching for food. These trails attract more ants, which increases the efficiency of food

collection [57, 58]. Similarly, the ACO algorithm uses simulated pheromones to identify promising paths in the search space for optimization problems. As the algorithm goes through iterations, pheromone levels rise on paths that lead to better solutions, guiding future search efforts.

The pheromone matrix stores pheromone levels along each path segment in two dimensions. In this method, $\tau_{ij}(t)$ represents the pheromone concentration between nodes i and j at time t , where t is the iteration count. Eventually, the pheromone evaporates. The pheromone matrix updates once all the ants have searched their paths. The global pheromone update rule is:

$$\tau_{ij}(t+1) = (1 - \rho)\tau_{ij}(t) + \Delta\tau_{ij}(t) \quad (6)$$

$$\Delta\tau_{ij}(t) = \sum_{k=1}^q \Delta\tau_{ij}^k(t) \quad (7)$$

$$\Delta\tau_{ij}^k(t) = \begin{cases} \frac{Q}{L_k}, & (i, j) \in \text{tour}_{\text{byant}k} \\ 0, & \text{otherwise} \end{cases} \quad (8)$$

where ρ ($0 < \rho < 1$) is the evaporation rate, q is the number of ants, Q is the pheromone amount, and L_k is the path length travelled by the k th ant during this phase. However, high pheromone levels may cause the algorithm to settle into a local optimum, while low concentrations may not attract other ants. This work constrains pheromone values using a max–min ant system (MMAS)-inspired method. Limiting pheromone values prevents search stagnation and improves algorithm convergence. ACO's k^{th} ant transition probability from node i to node j . Therefore, this work employs a method inspired by the max–min ant system (MMAS) to constrain the pheromone values, described as follows:

$$\tau_{ij}(t+1) = \begin{cases} \tau_{\max}, & \tau_{ij}(t+1) > \tau_{ij}(t+1) \\ \tau_{\max}(1 - \rho)\tau_{ij}(t) + \Delta\tau_{ij}(t), & \text{otherwise} \\ \tau_{\min}, & \tau_{ij}(t+1) < \tau_{\min} \end{cases} \quad (9)$$

$\tau_{\max}$ and $\tau_{\min}$ represent the maximum and minimum pheromone values. Limiting pheromone values prevents search stagnation and improves algorithm convergence. The ACO k th ant transition probability from node i to node j is:

$$P_{ij}^k = \begin{cases} \frac{(\tau_{ij}(t))^\alpha (\eta_{ij})^\beta}{\sum_{m \in U_k} (\tau_{im}(t))^\alpha (\eta_{im})^\beta}, & j \in U_k \\ 0, & \text{otherwise} \end{cases} \quad (10)$$

where U_k is the set of available next hop (NH) nodes for ants, η_{ij} is heuristic information, α is the pheromone parameter, and β is the heuristic information parameter. Ants follow Eq. (10) to the NH node and then the destination node. Once all q ants reach the destination node, the pheromone matrix is updated. This update reduces pheromone levels from evaporation, where ρ (from 0 to 1) is the parameter.

Evaporation prevents infinite pheromone concentrations. If ants don't choose a partial path, their pheromone concentration starts to drop, preventing them from taking this undesirable path. Ant pheromones gain value when left behind. Ant pheromones increase the likelihood of choosing better paths. We update the pheromone array values based on (7), (8), and (9). After updating the pheromone array, iteration continues. In this dynamic, ants choose paths based on selection criteria and update pheromone intensity after each path.

Algorithm 1 Ant Colony Optimization Algorithm

- Step 1: The sensors are dispersed at random throughout the implementation region.
- Step 2: Divide the distributed nodes into the most clusters possible.
- Step 3: ACO is used to create chains of clusters made up of all the clusters.
- Step 4: Through the use of ACO, a higher chain made up of all chain leads is created.
- Step.5. Data is transmitted through the cluster's chains to the chain leaderships, and then it is transmitted to the base station via the superior leader via the top chains.
- Step.6. If there are still any surviving nodes, continue to step 3; else, stop.

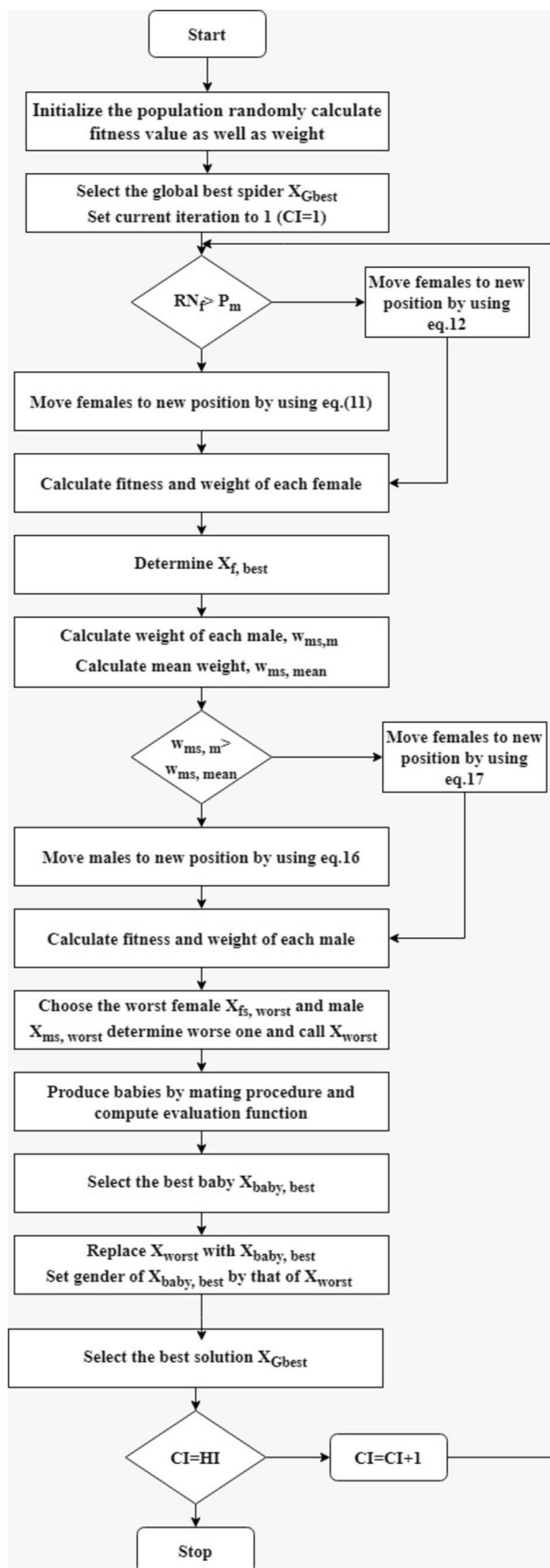


Fig. 2 Flowchart of proposed ISSO CH selection

Cluster formation in WSNs using ACO mimics ant foraging behavior to organize sensor nodes into clusters. Initially, each sensor node in the network is considered a CH candidate. Ants act as virtual agents, exploring the network based on pheromone trails and node visibility metrics. As ants move through the network, they evaluate nodes based on their pheromone levels (indicating attractiveness as CH) and other criteria such as node energy levels or proximity to other nodes. CHs are selected from nodes with higher pheromone levels and suitable attributes. These CHs then recruit neighboring nodes to form clusters, maximizing energy consumption, communication efficiency, and overall network performance. The process continuously updates pheromone trails to reflect the quality of cluster formations, ensuring adaptive and efficient organization tailored to WSNs' dynamic conditions.

3.4 CH selection using improved social spider optimization

The proposed ISSOA's flow diagram is depicted in Fig. 2. In the suggested system, the phases decide which CHs to transmit from different cluster members, whether directed towards the BS or through the CHs series. The network hosts can maintain a constant and dependable connection thanks to the choice of an ideal sensor, which helps CH transmit the data packets in an ordered manner. The cluster selection significantly influences the duration of WSNs, and the cluster centers are randomly selected based on one or more characteristics [59–61].

3.4.1 Initialization of spider population

Initial node placement in the targeted area is random. The position of the sensor network determines the exact location of the center node, which is immobile. The WSN suggests using the cluster approach to organize the node clusters, with the Cluster Centers chosen from each group. The best node is chosen to create the path from CH to the destination (i.e., BS). The selected node would send the data from one network to another if it required energy to do so; otherwise, it would select another node more appropriate for the data transmission. The high space of the web poses an optimization technique, which the solution space in SSA is intended to solution. All suggested solutions to the optimization original problem limit will be indicated by the spider's location on the web. As a result, the roles are revised with practical ideas to address the problem. When a spider moves, the adjacent web acts as a data transmission for the vibration. The answer is discovered using the food supplies at the various locations and each spider's ability to manipulate its position within the web. Because the spiders can only travel within the web

and cannot exit it, they are thought of as unrealistic optimization issue solutions. The SSO, like PSO, GA, and DE, is a metaheuristic algorithm; however, its configuration distinguishes it from these approaches. While PSO, GA, and DE typically generate new solutions only once, SSO produces three generations of new solutions. The first and second generations generate N_{fs} and N_{ms} new solutions, respectively. The third generation generates N new solutions, where N_{bs} is the number of baby spiders.

3.4.2 Movement of female spiders

The first generation generates N_{fs} solutions by moving all female spiders based on their positions. When the web vibrates, female spiders move first. Female spider f 's new position $X_{fs,f}$ depends on the vibration intensity and the positions of (1) the spider closest to them and (2) the best spider with the optimal weight. Therefore, depending on the movement probability P_m , Eqs. (11) or (12) determine female spider f 's new position. If a randomly generated number R_{nf} is greater than P_m and between 0 and 1, Eq. (11) moves spider f ; otherwise, Eq. (12) moves it.

$$X_{fs,f} = X_{fs,f} + rand.Vib_{nearest} \cdot (X_{nearest} - X_{fs,f}) + rand.Vib_{Gbest} \cdot (X_{Gbest} - X_{fs,f}) + rand.(rand - 0.5) \quad (11)$$

$$X_{fs,f} = X_{fs,f} - rand.Vib_{nearest} \cdot (X_{nearest} - X_{fs,f}) - rand.Vib_{Gbest} \cdot (X_{Gbest} - X_{fs,f}) + rand.(rand - 0.5) \quad (12)$$

Several symbols are defined in Eqs. (11) and (12): $Vib_{nearest}$ and $X_{nearest}$ refers to the spider closest to the female spider f , while Vib_{Gbest} and X_{Gbest} indicates the vibration intensity and position of the global best spider in the population. These expressions calculate spider vibrations:

$$Vib_{nearest} = w_{nearest} \cdot e^{-d_{f,nearest}^2} \quad (13)$$

$$Vib_{Gbest} = w_{Gbest} \cdot e^{-d_{f,Gbest}^2} \quad (14)$$

where $d_{f,nearest}$ and $d_{f,Gbest}$ are the distances between the female spider f and the nearest spider and the best spider, respectively. The weights of the nearest spider to f and the global best spider are $w_{nearest}$ and w_{Gbest} , respectively. The weight w_s of a spider is calculated by

$$w_s = \frac{FV_{Gworst} - FV_s}{FV_{Gworst} - FV_{Gbest}} \quad (15)$$

where, FV_{Gworst} and FV_{Gbest} represent the highest and lowest fitness values of the global worst and best spiders, respectively. FV_s represents the fitness value of the spider being considered.

3.4.3 Movement of male spiders

The male spiders move to new positions after the female spiders, generating N_{ms} new solutions. However, male spiders move differently from females. Equations (16) and (17) show that a male spider m new position depends on the nearest female spider or all male spiders. Similar to Eqs. (14) and (15), a condition determines which equation to use. The male spider m 's weight $w_{ms,m}$ is compared to the mean weight of all males $w_{ms,mean}$. If $w_{ms,m}$ is greater than $w_{ms,mean}$, male spider m will adjust its position using Eq. (16).

$$X_{ms,m} = X_{ms,m} + rand.Vib_{fs,nearest} \cdot (X_{fs,nearest} - X_{ms,m}) + rand.(rand - 0.5) \quad (16)$$

$$X_{ms,m} = X_{ms,m} + rand \cdot \left(\frac{\sum_{m=1}^{N_{ms}} X_{ms,m} \cdot w_{ms,m}}{\sum_{m=1}^{N_{ms}} w_{ms,m}} - X_{ms,m} \right) \quad (17)$$

Equation (16) determines the vibration intensity of the female spider closest to the male spider m to be $Vib_{fs,nearest}$

$$Vib_{fs,nearest} = w_{fs,nearest} \cdot e^{-d_{m,nearest}^2} \quad (18)$$

where $w_{fs,nearest}$ is the weight of the female spider closest to m and $d_{m,nearest}$ is their distance.

3.4.4 Mating to produce spiderlings

Male spiders interact with female spiders to maintain the spider community. This operation represents the third generation of new SSO solutions. Leading males mate with females within their designated zone. The zone is circular; its radius is determined by

$$\text{Radius} = \frac{\sum_{i=1}^D (Z_{i,max} - Z_{i,min})}{2 \cdot D} \quad (19)$$

where, $Z_{i,min}$ and $Z_{i,max}$ are the lower and upper bounds of the control variable Z , and D is its dimension. Spider babies are born after dominant males and females are established in the circle. The third generation of new solutions cannot be mathematically defined, unlike the first and second generations. Each new solution is generated by arbitrarily selecting control factors among these dominant males and females. Figure 2 shows the entire ISSOA search process for typical optimizations. The algorithm was designed to efficiently solve continuous space problems because spiders' webs have a constant solution. However, its evolutionary approach allows it to solve many problems perfectly. The sociable spider mimicked spider collaboration and possessed a unique evolutionary trait: the ability to mate with both sexes. Avoiding early solution convergence allows for an ongoing process that yields the best results.

Algorithm 2 CH selection using ISSO

Inputs: objective function $f(x)$, search the space's upper and lower limits $Z_{i,max}$ and $Z_{i,min}$, and set n spiders and T iterations;

Step 1: 1. Calculate the number of female and male spiders using formula (11), then initiate population sizes $X_{fs,f}$ ($i=1, 2, \dots, N_{fs}$), $X_{ms,m}$ ($i=1, 2, \dots, N_{ms}$) of different sexes. The problem dimension is D , fitness and weights are updated, and the mating radius is initialized using formula (19). with $t=0$

Step 2: While $t < T$

Step 3: Update female spider position using formulas (11) and (12).

Step 4: Update male spider position using formula (16).

Step 5 : Create new individuals using roulette;

Step 6: Update each individual's fitness value and weight;

Step 7: $t=t+1$

Step 8: end

Outputs: Find the best female and male spider colony individual and fitness value for CH selection

This work made substantial modifications to the traditional SSOA in the suggested algorithm and process for addressing the clustering problem in WSN using SSOA. The improved SSOA method (I-SSOA) mirrors the modifications discussed in these sections, including the initialization procedure, solution repair and reconfiguration for modeling sound waves through collective webs, and adaptive cooperative behavior in female ants, similar to the collective efficacy of male spiders. Any SSOA-solvable problem can have its binary form adjusted thanks to these four properties. The description of the fitness function taken into account for SSOA is given below:

3.4.4.1 Residual energy The CH gathers data from the conventional sensor network and transfers it to the BS, generating energy during the process. The CH requires a significant amount of energy to accomplish the tasks, prompting the completion of the previously listed tasks to replenish her energy reserves. As a result, the node with the highest energy, also known as CH f_1 , completes the tasks. According to the provided equation, the number (20) represents the remaining energy.

$$f_1 = \sum_{i=1}^m \frac{1}{E_{CH_i}} \quad (20)$$

3.4.4.2 The separation of the sensor network The transmission path formed from the nodes with the shortest data transmission from the BD determines how far the normal and Users are separated from the CH in the significant.

$$f_2 = \sum_{j=1}^m \frac{dis(s_i, CH_j)}{I_j} \quad (21)$$

According to Eq. (21), the sensor i and CH_j distance is represented as $dis(s_i, CH_j)$

3.4.4.3 Between BS-and-CH dimensions Amount of node energy consumed along the transmission channel determines the length between the BS and CH. As a result of the BS's far location from the CH, processing the data transfer demands more energy right away. The data is therefore transmitted via the shorter distance from the BS. Equation (22) calculates the objective function.

$$f_3 = \sum_{i=1}^m dis(CH_j, BS) \quad (22)$$

3.4.4.4 Node level Each CH has a corresponding number of sensor nodes, which determines the node degree. If there are fewer sensors present than the number of the chosen ones, CHs with a higher cluster member will have a quicker reduction in energy usage. As illustrated below, () is stated as a node degree in Eq. (23):

$$f_4 = \sum_{i=1}^m I_i \quad (23)$$

3.4.4.5 Centrality of nodes Equation (24) expresses the concept of "node centrality," referring to the distance between a node at the network's center and its neighboring nodes.

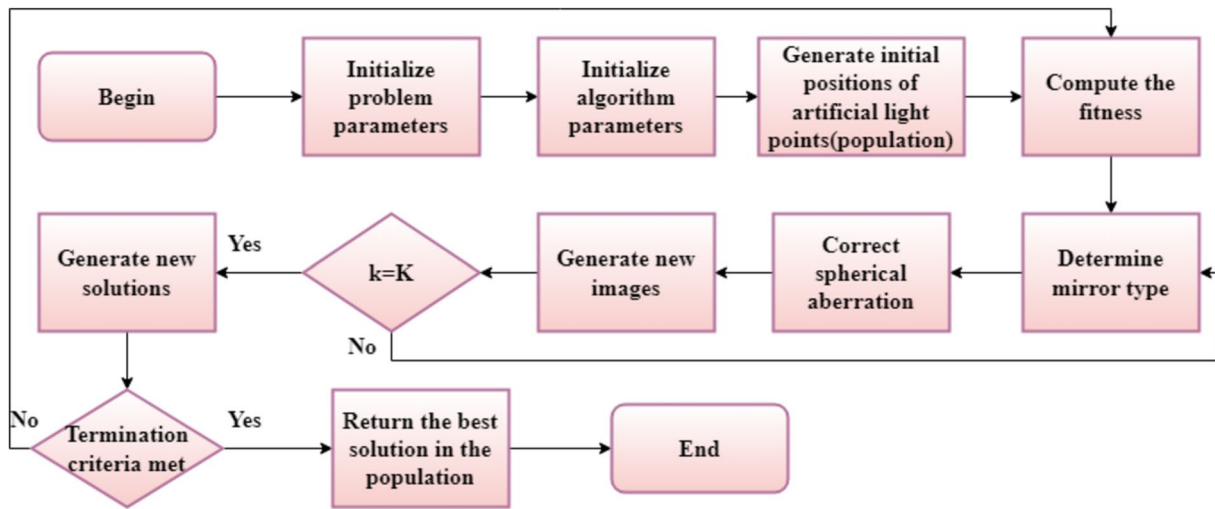


Fig. 3 Flowchart of OIO

$$f_5 = \sum_{i=1}^m \frac{\sqrt{\sum_{j \in n} \text{dist}^2(i, j) / n(i)}}{\text{Network dimension}} \quad (24)$$

The weighted values that are produced for each optimum solution are combined, and Eq. (25) is represented as follows:

$$\text{Fitness} = f_1 + f_2 + f_3 + f_4 + f_5 \quad (25)$$

Once the origin and destination networks establish a connection, they transfer data between them. The length among nodes, the based-on residual, and the optimization algorithm are all factors considered when constructing an optimal path using the objective functions. The supplier node sends data to the target node after creating the path connecting them. The range between nodes, the based-on residual between them, and the goal functions are all factors considered when constructing an optimal path.

3.5 Routing using optics inspired optimization algorithm

OIO is a new physics-inspired population-based metaheuristic algorithm. It is inspired by the optical properties of convex and concave mirrors, which solve complex problems. Light rays in a concave mirror converge toward the principal axis when reflected, whereas they diverge away from it in a convex mirror. These mirror effects influence OIO's exploration and exploitation abilities. Figure 3 depicts how new candidate solutions are generated, with each valley symbolizing a concave mirror and each peak representing a convex mirror. The primary goal of the multi-hop routing problem is to reduce network energy consumption while maintaining

balanced loads on CHs [62, 63]. The first objective, g_1 , was to maximize network longevity by selecting the NH with the highest residual energy to forward data packets. The second objective function, g_2 , minimizes the distance from CHs to the NH and then to the BS. For a maximization problem, the reciprocal of the distance minimizes network energy consumption. To balance cluster load distribution, the final objective function G_3 minimizes node degree (reciprocal chosen for maximization). Let a_{ij} represent a boolean variable described as follows:

$$a_{ij} = \begin{cases} 1 & \text{if } NH(CH_i) = CH_j, \forall i, j \leq m \\ 0 & \text{otherwise} \end{cases} \quad (26)$$

$$\text{Maximize : } F = \varphi_1 \times g_1 + \varphi_2 \times g_2 + \varphi_3 \times g_3$$

Subject to:

$$\text{dis}(CH_i, NH) \times a_{ij} \leq d_{\max}, \forall_{NH} \in \{C + BS\} \quad (27a)$$

$$\sum_{j=1}^m a_{ij} = 1 \text{ and } i \neq j \quad (27b)$$

$$0 < \varphi_1, \varphi_2, \varphi_3 < 1 \quad (27c)$$

According to Eq. (27a), the NH must be within the communication range of CH_i . In Eq. 27b, the second constraint mandates that each CH_j delivers data to a unique NH. Equation (27c)'s final constraint requires control parameter values (weights) to be between 0 and 1. During routing, each artificial light point's dimension matches CHs.

Let $O_i = (O_i, O_{i,2}, \dots, O_{i,m})$ denote the i th artificial light point, where each $O_{i,d} \in (0,1)$, $\forall_i 1 \leq i \leq N_p$, $\forall_d 1 \leq d \leq m$, is randomly initialized by a real number w ranging between 0 and 1. Subsequently, a novel mapping function is employed to determine the next-hop direction towards the BS. This mapping function is represented by Eq. (28), where C_r denotes the number of nodes within its communication range and X_{ij} represents a real number.

$$f(x) = \begin{cases} i, & \text{for which } \left(\frac{i}{C_r} - X_{i,j}\right) \text{ is minimum,} \\ \forall_i, 1 \leq i \leq C_r \end{cases} \quad (28)$$

The main goal is to find the best route from each CH to the BS.. This is accomplished by defining a fitness function that considers distance, residual energy and node degree.

Algorithm 3 OIO Routing

Input: Set of chosen CH nodes $CH = (CH_1, CH_2, CH_3, \dots, CH_m)$,
 Predefined number of artificial light points N_p , and their dimension (m).
 Step 1: Initialize the artificial light points $O_{i,d} = (0, 1), \forall_i 1 \leq i \leq N_p$
 Step 2: **While** $i \neq N_p$ **do**
 Find fitness O_i using equation 32.
 end
While iteration! = Max_iteration **do**
 While $i \neq N_p$ **do**
 Use roulette wheel selection to select population member j as F_j .
 if (fitness (O_i) > fitness (F_j))
 (1) Concave mirror
 (2) Center of curvature calculation
 (3) Artificial light point calculation
 Else
 (1) Convex mirror
 (2) Calculate radius of curvature
 (3) Determine place of artificial light point
 end
 if (fitness (New_i) > fitness (O_i)) **then**
 Replace i^{th} population member with New_i
 end
 end
end
 Step 3: CH chooses NH for transmission of data using max (fitness (O_i))
 Output: Optimal routing path for each CH

Table 2 Simulation parameters setup

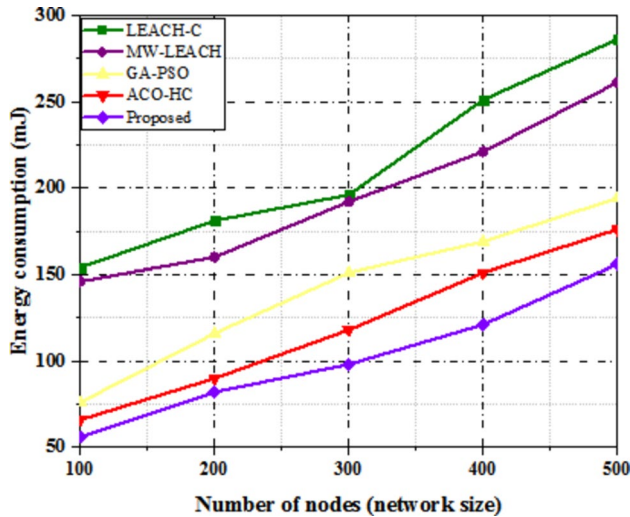
Parameter	Value
Bandwidth	20 kbps
Number of nodes	500
Deployment area	500 m × 500 m
Initial energy	0.1 J
Packet sending rate	1 packet/sec
Packet size	512 bytes
Node distribution	Random

3.5.1 Energy remaining in the next-hop node

During data transmission, each CH chooses a closer upcoming node to send its data packets to the BS. The chosen next-hop node is also required to fulfil its duties, the chief among

Table 3 Energy Consumption (mJ)

No. of nodes	LEACH-C	MW-LEACH	GA-PSO	ACO-HC	Proposed
100	154	146	76	66	56
200	181	160	116	90	82
300	196	192	151	118	98
400	251	221	169	151	121
500	286	261	194	176	156

**Fig. 4** Energy consumption

which is to convey gathered data from inter and intra-membership nodes following aggregation. Consequently, just the next node with more incredible energy remaining is a preferable option. As a result, g_1 is maximized according to the first objective connected to residual energy:

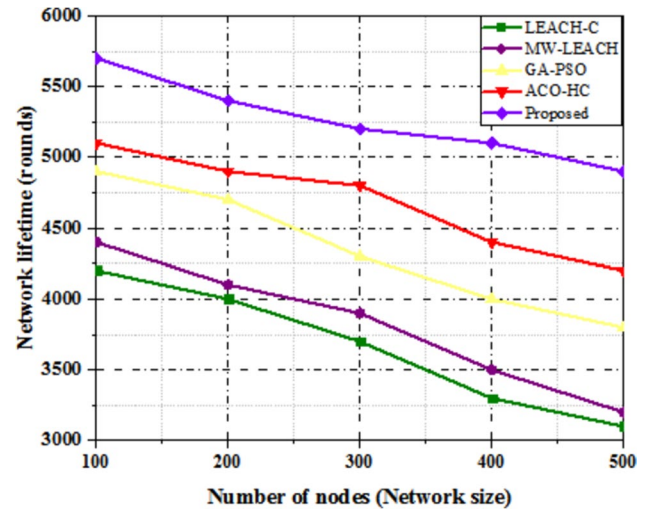
$$\text{Maximize } g_1 = \sum_{i=1}^m ECH_i \quad (29)$$

3.5.2 Distance

The CH must travel this Euclidean distance in order to reach the BS from the next-hop node. Transmission distance affects energy use. It utilizes the least power when it's at its lowest. The secondary target is to shorten the length from CH to the following hop and from the following hop. As a result, a second goal related to distances is the maximization of g_2 .

Table 4 NLT (Rounds)

No. of nodes	LEACH-C	MW-LEACH	GA-PSO	ACO-HC	Proposed
100	4200	4400	4900	5100	5700
200	4000	4100	4700	4900	5400
300	3700	3900	4300	4800	5200
400	3300	3500	4000	4400	5100
500	3100	3200	3800	4200	4900

**Fig. 5** Network lifetime

$$\text{Maximize } g_2 = \frac{1}{\sum_{i=1}^m \text{dis}(CH_i, NH) + \text{dis}(NH, BS)} \quad (30)$$

3.5.3 Node level

It is the number of individuals who are not CH members but are yet members of a specific CH. Node degree is an essential element in the next-hop selection process. The next hop will receive less info from its members and take longer to complete tasks if it is chosen with a lower node degree.

$$\text{Maximize } g_3 = \frac{1}{\sum_{i=1}^m I_{mi}} \quad (31)$$

The normalization function is used to bring all of the aforementioned objectives' range values into the same range. The aims are not in contradiction with one another here, as well. The weighted sum technique is so appropriate. Each objective's weight is denoted by the symbols $\varphi_1, \varphi_2, \varphi_3$

$$\text{Maximize Fitness} = \varphi_1 \times g_1 + \varphi_2 \times g_2 + \varphi_3 \times g_3 \quad (32)$$

Where $\sum_{i=1}^3 \varphi_i = 1, \varphi_i \in (0,1)$

OIO algorithm leverages local search strategies where sensor nodes explore nearby neighbors to gather information about potential routing paths. This local exploration phase resembles how light propagates through optical media, with nodes assessing connectivity based on factors like signal strength and node proximity. Clustering plays a pivotal role as nodes self-organize into clusters, facilitating intra-cluster communication and reducing the complexity of path selection. The algorithm integrates objective functions that prioritize metrics such as energy efficiency, latency reduction, and data reliability. Through iterative refinement and global optimization, OIO algorithm dynamically adapts to changing network conditions, ensuring robust and adaptive routing paths that optimize network performance and extend the network's operational lifespan.

4 Results analysis

The MATLAB simulation tool executes the proposed algorithm. The system's domain organizes nodes in WSN according to a standard transmission protocol. The process of delivering packets to the system domain starts with a random selection of a point on the system's perimeter. We compare the results of the proposed method with those of five established solutions: LEACH-C, MW-LEACH, GA-PSO, and ACO-HC. For 500 nodes, the proposed method improves energy consumption, PDR, E2ED, NLT, throughput, jitter, buffer occupancy, and PLR (simulation results). The proposed simulation parameters are in Table 2. The dimensions for deployment areas are typically measured in meters (m) unless stated otherwise. Thus, a deployment area of 500×500 indicates that the sensor nodes are distributed across a 500-m by 500-m region. Several WSN nodes are active or inactive. Node ranges and proximity-based clustering are defined by the effective clustering method's periodic monitoring.

4.1 Energy consumption

The longevity of WSNs is influenced by their energy consumption. This energy efficiency is quantified by combining the transmitted energy E_{tx} and the received energy E_{rx} across all nodes.

$$E_{\text{consumption}} = (E_{rx} \times N) + E_{tx} \quad (33)$$

where E_{rx} denotes the received energy and E_{tx} represents the transmitted energy, while N stands for the number of nodes.

Table 3 and Fig. 4 illustrate the energy performance. The suggested (ACO-ISSOA) method consumes less energy than other routing protocols for LEACH-C, ACOHC, GA-PSO, and MW-LEACH. The energy cost ranges from 0 to 300. The proposed algorithm utilizes 56 mJ of energy at 100 nodes, which is less than numerous approaches. When compared with other approaches, the suggested technique uses more energy because of the increase in node count. LEACH-C, ACOHC, GA-PSO, and MW-LEACH are the techniques available, each consuming 154 mJ, 146 mJ, 76 mJ, and 66 mJ of energy at 100 nodes, respectively.

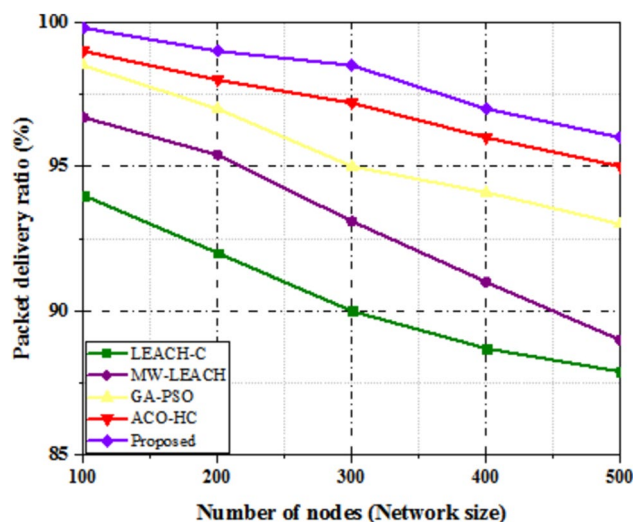


Fig. 6 PDR

Table 5 PDR (%)

No. of nodes	LEACH-C	MW-LEACH	GA-PSO	ACO-HC	Proposed
100	94	96.7	98.5	99	99.5
200	92	95.4	97	98	99
300	90	93.1	95	97.2	98.5
400	88.7	91	94.1	96	98
500	87.9	89	93	95	97

Table 6 E2ED (sec)

No. of nodes	LEACH-C	MW-LEACH	GA-PSO	ACO-HC	Proposed
100	7.1	6.8	5.2	3.9	3.3
200	8.3	7.6	6.3	5.0	4.5
300	9.8	8.4	7.5	6.4	5.7
400	10.5	9.5	9.2	7.8	6.4
500	11.9	10.6	10.4	9.0	7.3

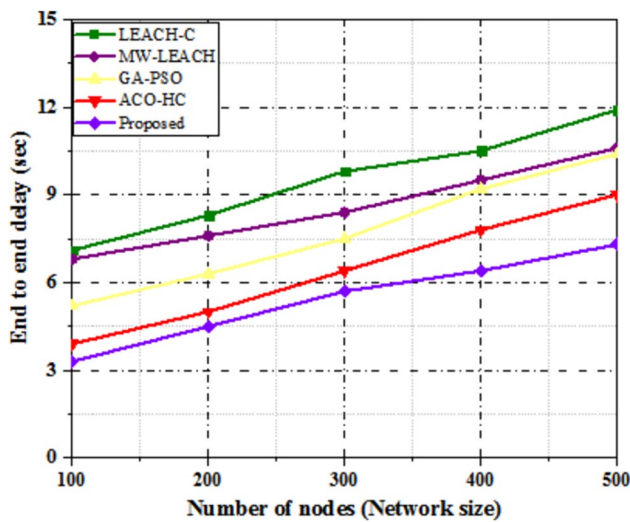


Fig. 7 E2ED

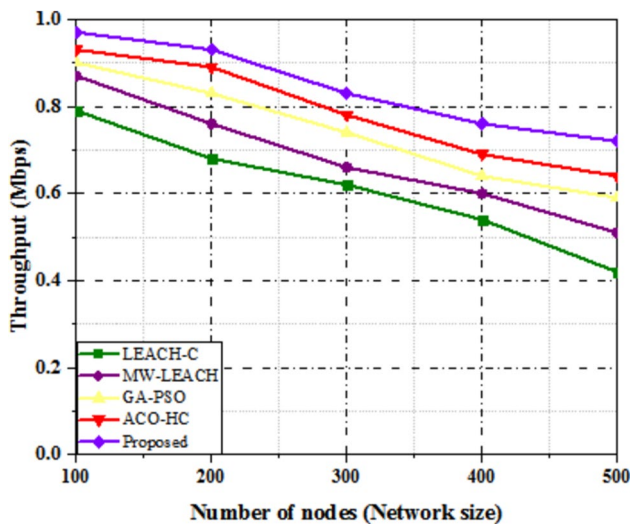


Fig. 8 Throughput

Table 7 Throughput (Mbps)

No. of nodes	LEACH-C	MW-LEACH	GA-PSO	ACO-HC	Proposed
100	0.79	0.87	0.90	0.93	0.99
200	0.68	0.76	0.83	0.89	0.93
300	0.62	0.66	0.74	0.78	0.83
400	0.54	0.60	0.64	0.69	0.76
500	0.42	0.51	0.59	0.64	0.72

With 200 nodes, the proposed approach consumed 82 mJ, while LEACH-C, MW-LEACH, GA-PSO, and ACO-HC used 181, 160, 116, and 90 mJ, respectively. For 300 nodes, the introduced approach consumed 98 mJ,

Table 8 PLR (%)

No. of nodes	LEACH-C	MW-LEACH	GA-PSO	ACO-HC	Proposed
100	10	9	5	3	1
200	14	11	6	5	3
300	15	13	8	6	4
400	18	14	9	7	6
500	19	15	10	8	7

while LEACH-C, MW-LEACH, GA-PSO, and ACO-HC used 196, 192, 151, and 118 mJ, respectively. With 400 nodes, the hybrid ACO-ISSOA method consumed 121 mJ, while LEACH-C, MW-LEACH, GA-PSO, and ACO-HC used 251, 221, 169, and 151 mJ, respectively. Thus, the hybrid approach transmits data efficiently.

4.2 NLT

Network lifetime refers to the duration from the position of the sensor nodes to the eventual failure of the network. When the initial sensor becomes nonfunctional, coverage diminishes, network partitions occur, and the percentage of sensor nodes that cease functioning increases.

$$E[L_{\text{lifetime}}] = \frac{\sigma_0 - E_{\text{estimate}}[E_{RE}]}{CP + \vartheta E[E_{CE}]} \quad (34)$$

In the equation above, CP represents the constant power consumption across the entire network, $E_{\text{estimate}}[E_{RE}]$ denotes the estimated remaining energy not used in processing, ϑ stands for the average sensor reporting rate indicating the data collected per unit time, and the initial non-rechargeable energy is denoted by σ_0 . Traditional methods have a shorter network lifetime than the proposed method.

Table 4 and Fig. 5 compare network lifetimes. WSN packets should travel farther and use more energy. Choosing the best CHs with a hybrid ACO-ISSOA method and finding the shortest path efficiently in OIO both lower the amount of energy that CHs use. Energy demand reduction extends the network's lifespan. The graph shows that the new ACO-ISSOA method has a longer network lifetime than other methods. As nodes increase, network lifespan decreases. The new method uses 5700 rounds and 100 nodes, which is longer than existing methods. The current algorithms LEACH, MW-LEACH, GA-PSO, and ACOHC have network lifetimes of 4200, 4400, 4900, and 5100 rounds for 100 nodes.

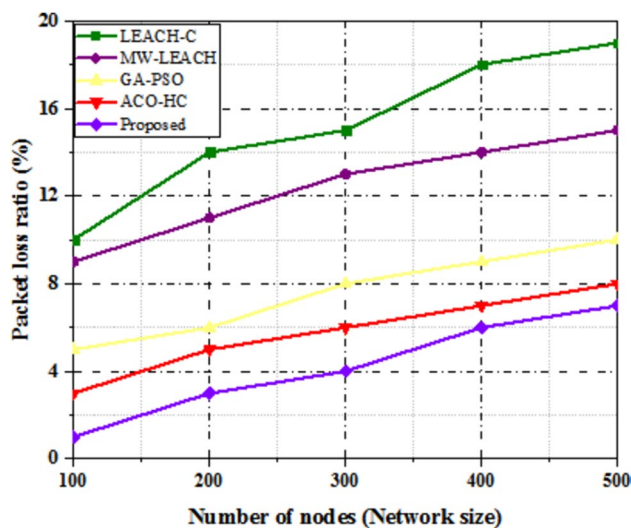


Fig. 9 PLR

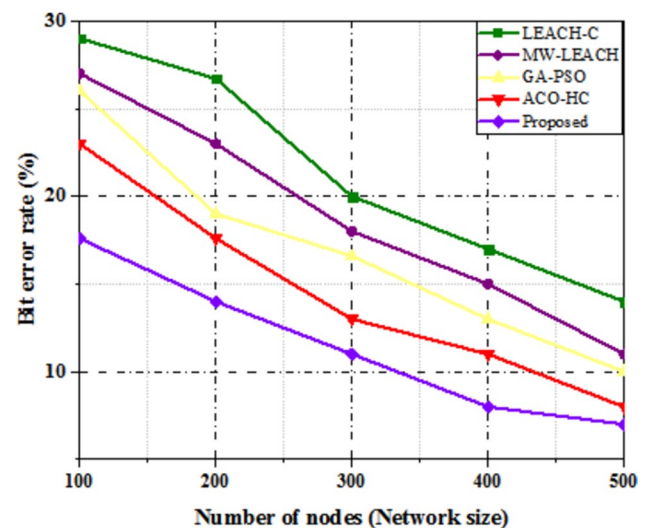


Fig. 10 Bit error rate

Table 9 BER (%)

No. of nodes	LEACH-C	MW-LEACH	GA-PSO	ACO-HC	Proposed
100	29	27	26	23	17.6
200	26.7	23	19	17.6	14
300	20	18	16.6	13	11
400	17	15	13	11	8
500	14	11	10	8	7

Table 10 Channel load (%)

No. of nodes	LEACH-C	MW-LEACH	GA-PSO	ACO-HC	Proposed
100	22.5	24	26	27	29
200	18	20	21.6	24	26
300	15.6	18	20	22	25
400	13	16	17	19	23
500	11	14	16	16.6	22

4.3 PDR

The cluster-based routing scheme prioritizes CHs based on residual energy and routes via the shortest path, improving packet forwarding reliability and efficiency. This method reduces packet retransmissions and energy usage. Simulations show that PDR decreases with node count. PDR is the ratio of data packets received by the recipient to those sent by the sender.

$$PDR = \frac{\text{Total number of data packets received by the receiver}}{\text{Total number of data packets transmitted by the transmitter}} \quad (35)$$

Table 5 and Fig. 6 illustrate the performance of the newly introduced ACO-ISSOA approach and the current method in terms of PDR. Simulations demonstrate that the PDR falls as the number of nodes rises. Comparing the newly introduced method to existing ones shows a higher packet delivery rate of 99.8% across 100 nodes. The proposed approach is more effective than the current approaches. The current methods, LEACH-C, MW-LEACH, GA-PSO, and ACOHC, have packet delivery rates of 94%, 96.7%, 98.5%, and 99%, respectively, out of 100 nodes. A high PDR is crucial for

effective data transmission. When PDR is high, all data reaches the destination node without loss. Increasing the number of nodes significantly boosts PDR, thereby enhancing the performance and efficiency of the proposed model compared to current methods.

4.4 E2ED

The proposed OIO method's latency is minimized through efficient delivery probability and shortest path determination calculations. OIO seeks to alleviate network congestion. Routing data from source to destination takes on average seconds. This average delay transmits data between network nodes from the initial node to the destination node.

$$\text{Average packet delay} = \frac{\text{Packet received time}}{\text{packet transmitted time}} \quad (36)$$

Table 6 and Fig. 7 compare E2ED for both the suggested (ACO-ISSOA) and existing methods. At 100 nodes, the proposed method has a lower E2ED (3.3 s) than other methods. At 100 nodes, LEACH-C, MW-LEACH, GA-PSO,

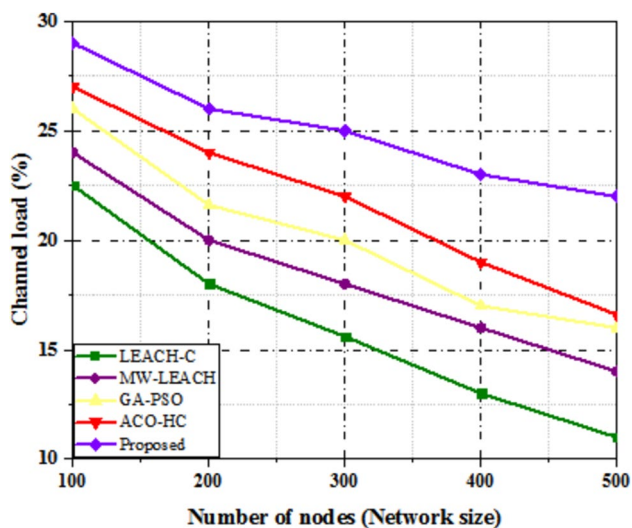


Fig. 11 Channel load

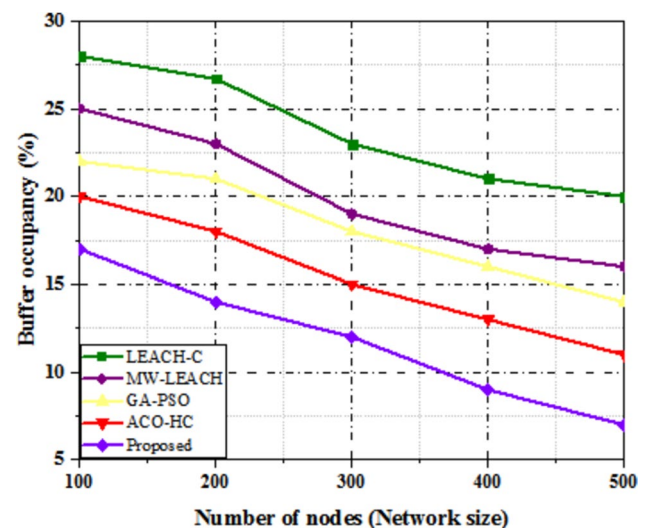


Fig. 12 Buffer occupancy

Table 11 Buffer occupancy (%)

No. of nodes	LEACH-C	MW-LEACH	GA-PSO	ACO-HC	Proposed
100	28	25	22	20	17
200	26.7	23	21	18	14
300	23	19	18	15	12
400	21	17	16	13	9
500	20	16	14	11	7

and ACOHC have E2EDs of 7.1, 6.8, 5.2, and 3.9 s, respectively. To improve link reliability and packet delays, the OIO method finds the shortest path.

4.5 Throughput

Throughput is defined as the data rate at which information is successfully transmitted across a network, which reflects the time it takes for a packet to travel from sender to receiver.

$$\text{Throughput} = \frac{\text{total amount of packets sent}}{\text{Time required for transmitting the packet}} \quad (37)$$

Figure 8 and Table 7 display the throughput results. The figure clearly shows a significant improvement in the proposed (ACO-SSOA) method. The proposed approach achieves a higher throughput (0.99 Mbps) on 100 nodes than other approaches. As the number of nodes increased, throughput decreased. We discovered that the introduced throughput significantly outperforms other energy-saving techniques. The current methods, LEACH-C, MW-LEACH, GA-PSO, and ACOHC, have throughputs of 0.79 Mbps, 0.87 Mbps,

Table 12 Jitter

No. of nodes	LEACH-C	MW-LEACH	GA-PSO	ACO-HC	Proposed
100	0.9	0.8	0.76	0.67	0.50
200	0.80	0.77	0.73	0.58	0.48
300	0.76	0.72	0.71	0.58	0.47
400	0.7	0.66	0.62	0.56	0.45
500	0.64	0.6	0.53	0.50	0.43

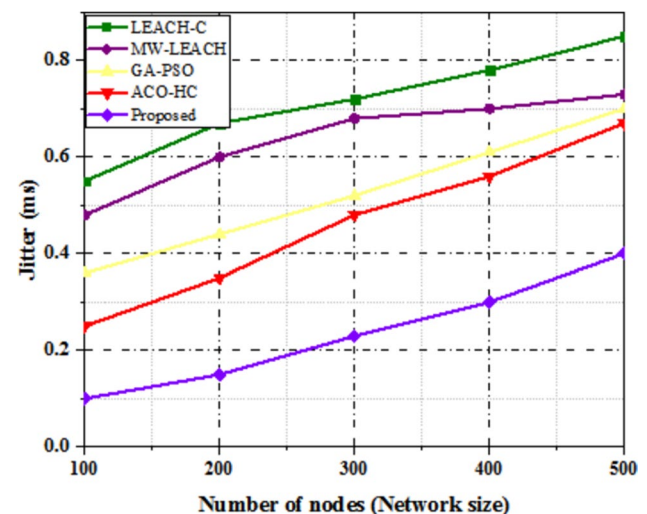
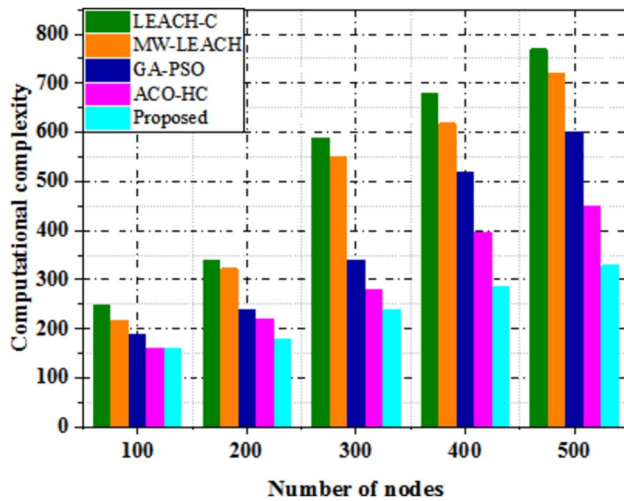


Fig. 13 Jitter

0.90 Mbps, and 0.93 Mbps, respectively, at 100 nodes. We select CH using an appropriate fitness function, which employs ISSOA. This clustering algorithm has demonstrated excellent outcomes. The lifespan and residual energy of the

Table 13 Computational complexity

No. of nodes	LEACH-C	MW-LEACH	GA-PSO	ACO-HC	Proposed
100	248	217	188	160	159
200	339	322	240	220	180
300	588	550	340	280	240
400	680	618	520	397	285
500	769	720	600	450	329

**Fig. 14** Computational Complexity

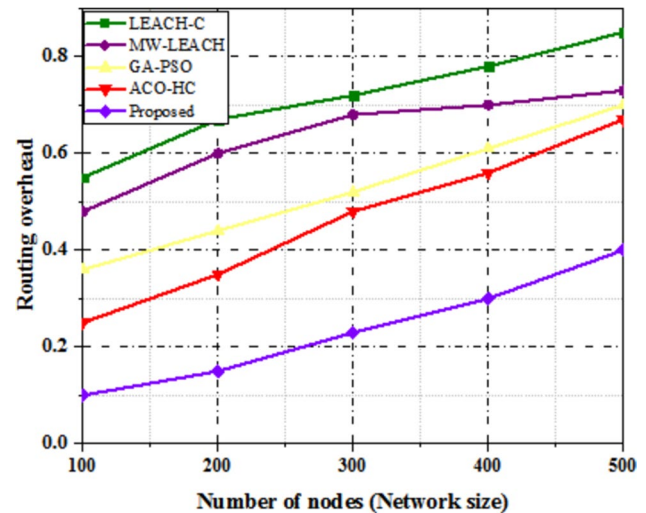
WSN closely correlate with the number of received data packets. Increasing the number of nodes typically decreases throughput.

4.6 PLR

Packet loss occurs when data transmitted from a transmitter node does not reach the receiver node. Table 8 shows the packet loss ratios for both the conventional and introduced hybrid approaches. The hybrid method proposed here has a lower packet loss ratio than previous solutions, emphasizing the efficiency of the introduced OIO during data transfer. The graph shows a significant reduction in packet loss with the proposed model compared to existing methods. Figure 9 presents a PLR comparison between the recently introduced (ACO-ISSOA) algorithm and other existing algorithms. The results show that the new approach outperforms the alternatives considered. In comparison to other methods that are already in use, the newly introduced method achieves a lower PLR at 100 nodes. As the number of nodes increases, the rate of packet loss rises. The current methods, LEACH-C, MW-LEACH, GA-PSO, and ACOHC, have packet loss rates of 10%, 9%, 5%, and 3%, respectively, at 100 nodes.

Table 14 Routing overhead

Number of nodes	LEACH-C	MW-LEACH	GA-PSO	ACO-HC	Proposed
100	0.55	0.48	0.36	0.25	0.10
200	0.67	0.60	0.44	0.35	0.15
300	0.72	0.68	0.52	0.48	0.23
400	0.78	0.70	0.61	0.56	0.30
500	0.85	0.73	0.70	0.67	0.40

**Fig. 15** Routing overhead

The introduced method offers significant savings compared to other energy-saving techniques.

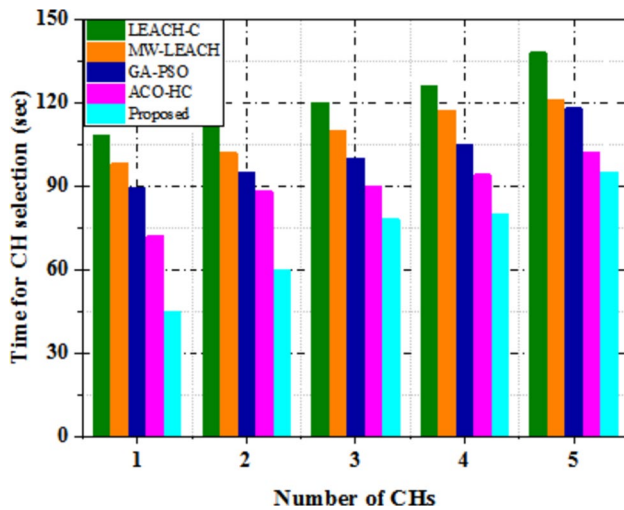
4.7 Bit error rate (BER)

In order to determine the dependability and efficiency of communication links, which are necessary for effective data aggregation, routing decisions, and overall network performance, BER analysis is an important aspect. BER analysis helps to fine-tune parameters such as transmission power and error correction mechanisms in WSNs to achieve the highest levels of communication reliability and energy efficiency.

Table 9 and Fig. 10 display the BER performance of the proposed solution (ACO-ISSOA) using current techniques. Comparing the introduced protocol to other existing protocols, simulations show that it achieves a low BER (17.6%) at 100 nodes. The BER increases with the number of nodes in a network. As a result, the newly introduced method outperforms the existing algorithms. The BERs for the existing methods LEACH-C, MW-LEACH, GA-PSO, and ACOHC are 29%, 27%, 26%, and 23%, respectively, at 100 nodes.

Table 15 Time for CH selection (sec)

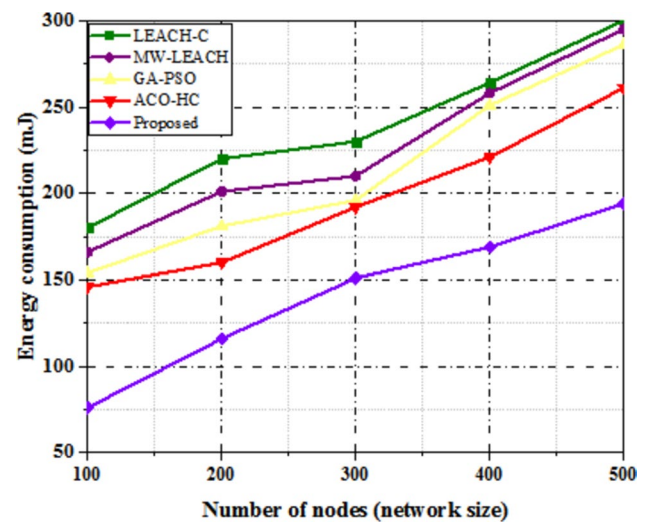
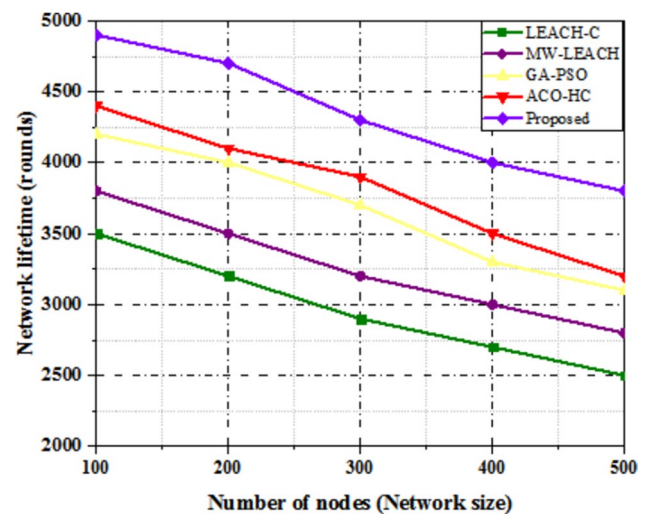
Number of cluster heads	LEACH-C	MW-LEACH	GA-PSO	ACO-HC	Proposed
1	108	98	89	72	45
2	115	102	95	88	60
3	120	110	100	90	78
4	126	117	105	94	80
5	138	121	118	102	95

**Fig. 16** Time for CH selection

4.8 Channel load

Channel load analysis involves assessing the utilization or congestion levels of communication channels in the network. This analysis is essential for enhancing network performance, ensuring effective data transmission, and preventing packet loss caused by channel saturation. As the network grows or becomes denser with additional sensor nodes, the overall traffic load on channels naturally increases. Proposed methods tailored for scalable networks are capable of accommodating higher traffic volumes, potentially leading to increased channel loads compared to algorithms designed for smaller-scale deployments.

Table 10 and Fig. 11 show the channel loading comparison between the newly introduced (ACO-ISSOA) and existing methods. As the number of nodes rises, the graph demonstrates that the channel load decreases. Compared to other existing protocols, the implementation achieves a high channel load (29%) at 100 nodes. The current methods LEACH-C, MW-LEACH, GA-PSO, and ACOHC, have channel loads of 22.5%, 24%, 26%, and 27%, respectively, at 100 nodes.

**Fig. 17** Energy consumption- BS outside sensors**Fig. 18** NLT analyses—BS outside sensors

4.9 Buffer occupancy (%)

Buffer occupancy analysis is centered on assessing how buffers at sensor nodes are utilized, storing both incoming and outgoing data packets awaiting transmission. This evaluation is critical for evaluating the effectiveness of routing protocols and ensuring efficient data management across the network. Typically, buffer occupancy analysis entails measuring parameters like average buffer size, packet arrival rates, packet transmission rates, and queuing delays to gauge the efficiency and management of buffers.

Table 11 and Fig. 12 show the buffer occupancy performance of the newly introduced ACO-ISSOA and traditional methods, respectively. Simulations show that the newly introduced protocol achieves low (17%) buffer occupancy

Table 16 Comparative analysis of the proposed method with optimization algorithm using 100 nodes

Algorithms	Throughput (%)	Delay (ms)	PDR (%)	Communication cost (j)	Energy consumption (J)	Computational time (ms)
LEACH-C	85.7	0.0049	89	0.30	0.50	4.8
PSO	89.5	0.0045	90.2	0.29	0.48	4.5
ACO	90.1	0.0040	92	0.25	0.40	3.9
GA-PO	92.7	0.0038	93.5	0.20	0.35	3.5
ASSO	93.2	0.0035	94	0.19	0.28	3.36
ACO-HC	94.5	0.0030	95	0.16	0.25	3.1
CRWO	95.7	0.0029	96.3	0.14	0.22	2.9
Proposed	99.0	0.0018	99.5	0.008	0.16	1.5

at 100 nodes compared to other existing protocols. As the number of nodes increases, buffer occupancy decreases. Therefore, compared to other energy-saving protocols, the present protocol is more effective. Buffer occupancy rates for the current methods (LEACH-C, MW-LEACH, GA-PSO, and ACO-HC) are 28%, 25%, 22%, and 20%, respectively.

4.9.1 Jitter

Jitter analysis assesses the variability in packet arrival times received at the endpoint. It quantifies the irregularity or inconsistency in packet delay, which can impact the quality of real-time applications like synchronization of sensor data. Jitter analysis is essential for evaluating the effectiveness of clustering and routing protocols in WSNs, as it directly influences the reliability and timeliness of data transmission.

Table 12 and Fig. 13 present a comparison of the jitter levels of the proposed (ACO-ISSOA) protocol with those of the existing protocols, respectively. When compared to other protocols that are currently in use, the findings indicate that the proposal is capable of achieving a low jitter value of 0.50 ms at 100 nodes. Jitter values continue to decrease as the number of nodes in the network increases. Consequently, the recently implemented protocol outperforms the various other energy sources currently in use. Currently available protocols, namely LEACH-C, MW-LEACH, GA-PSO, and ACOHC, have jitter values of 0.9 ms, 0.8 ms, 0.76 ms, and 0.67 ms, respectively.

4.9.2 Computational complexity

We determined whether the proposed method, along with other existing methods like LEACH-C, ACOHC, GA-PSO, and MW-LEACH, had a higher level of computational complexity. A significant increase in the number of nodes results in a greater complexity. Table 13 shows the computational complexity performance, and Fig. 14 shows that the proposed method uses less computational time than conventional algorithms.

Existing algorithms often underperform due to improper consideration of fitness functions during CH selection. In the ACO-ISSOA clustering approach, diverse fitness variables are utilized to identify suitable CHs among the sensors. Subsequently, an optimized route creation using OIO is employed to minimize node energy depletion. Consequently, compared to existing methods, the introduced hybrid ACO-ISSOA approach achieves a longer network lifespan, attributed to the increased volume of data packets transmitted to the BS.

The computational complexity of proposed ACO-ISSO is,

$$O(k_{ACO} \cdot m_{ACO} \cdot n_{ACO} + n_{ISSOA} \cdot m_{ISSOA} + N_{OIO} \cdot M_{OIO}).$$

This hybrid approach may reduce computational overhead compared to GA-PSO and ACOHC while optimizing WSN energy usage and network reliability.

4.9.3 Routing overhead

The performance of routing overhead between the proposed ACO-ISSOA method and existing methodologies (LEACH-C, ACOHC, GA-PSO, and MW-LEACH) was compared. Table 14 presents the evaluations of these existing methods. Figure 15 illustrates that the proposed ACO-ISSOA achieves superior results compared to the existing approaches. Specifically, ACO-ISSOA demonstrates reduced routing overhead compared to LEACH-C, ACOHC, GA-PSO, and MW-LEACH. The figure indicates that the proposed method achieves a lower routing overhead of 0.10 at 100 nodes, which is significantly better than LEACH-C (0.55), MW-LEACH (0.48), GA-PSO (0.36), and ACOHC (0.25).

4.9.4 Time for CH selection

Table 15 and Fig. 16 present the CH selection times, respectively. The proposed method outperformed the previously used methods in terms of execution times. As the number of CHs increased, the execution time also increased.

Table 17 Statistical analysis of residual energy

Algorithm	Samples (N)	Mean	Standard deviation	Standard error
LEACH-C	50	0.0892	0.083	0.043
MW-LEACH	50	1.6483	0.42	0.165
GA-PSO	50	−0.0532	0.049	0.037
ACO-HC	50	−0.0451	0.037	0.052
Hybrid ACO- ISSOA	50	1.9217	0.558	0.045

Specifically, the introduced approach achieved a low execution time of 45 s for the initial CHs, whereas existing methods such as LEACH-C, MW-LEACH, GA-PSO, and ACOHC required 108, 98, 89, and 72 s, respectively.

4.9.5 Network performance analysis- placement of the base station (BS) outside sensors

It assesses the network's base station (BS) placement outside sensors. Traffic to the base station (BS) greatly impacts network performance. Figure 17 compares BS placement algorithms for outside sensors. The proposed algorithms use less energy. The estimated network lifetime is defined as the time between sensor node activation and network shutdown. Figure 18 compares the suggested model's network lifetime to conventional approaches where the BS is located outside the sensor network. Data transmission increases as the network ages. Using the same number of nodes, the proposed method achieves 4900 rounds of network lifetime with the BS outside the sensor network.

BS distribution among sensors reduces energy consumption, balances network load, and extends the network's lifespan. The proposed algorithm uses the least energy and lasts the longest, regardless of BS placement.

4.9.6 Performance analysis with other optimization algorithm

Table 16 compares the proposed ACO-ISSO algorithm with a conventional algorithm involving 100 nodes. These parameters include energy consumption, PDR, E2D communication cost, throughput, and computational time. The LEACH-C model, for example, had a minimum PDR of 89%, 85.7%

throughput, and a communication cost of 0.30j. Similarly, PSO, ACO, GA-PSO, ASSO, ACO-HC, and CRWO demonstrated comparable performance metrics. In contrast, the proposed model performed better with a higher PDR of 98%, throughput of 99.5%, a shorter delay of 0.0018 ms, and a lower communication cost of 0.008 j.

4.9.7 Statistical analysis

Table 17 shows the residual energy comparison of the recommended ACO-ISSOA scheme with other algorithms. After analysis, the hybrid ACO-ISSOA scheme had the highest average score of 1.9217. In Table 18, the analysis of variance (ANOVA) test shows a p-value of 0.002 (less than 0.005). The ACO-ISSOA scheme outperforms benchmark algorithms significantly.

5 Conclusion

This research work describes hybrid ISSOA-ACO-based energy-efficient clustering and an OIO-based routing system for WSNs. It introduces energy-efficient clustering based on a hybrid ACO with an additional ISSOA. We choose a CH from the appropriate nodes based on factors like residual energy, QoS impact, and node location, each of which has multiple sub-criteria. Additionally, using the CH strategy reduces traffic and conserves energy. Finally, we use a routing protocol known as OIO. Using MATLAB simulation tools, we analyze the data gathered from real-time data transfer situations. After comparing the proposed method to other algorithms like LEACH-C, MW-LEACH, hybrid GA-PSO, and ACOHC, it becomes clear that it works better. Specifically, the ACO-ISSOA protocol extends network lifetime by 35.71%, 29.54%, 16.32%, and 11.76% in comparison to the LEACH-C, MW-LEACH, GA-PSO, and ACO-HC protocols, respectively. Finding routes that maximize the remaining energy for potential multi-hop cross-communications and adding more base stations nearby to lessen the burden on cluster heads will increase the network's overall energy efficiency and lifespan. To enhance WSN performance, future research will propose innovative deep learning and efficient data prediction-based clustering and routing methods.

Table 18 ANOVA test-result

Description	Sum of square	df	Mean square	F-value	Significance
Between groups	133.428	8	28.382	819.59	0.002
Within groups	5.8347	142	0.0454	—	—
Total	139.2627	150	—	—	—

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Declarations

Conflicts of interest The authors declare no conflict of interest.

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